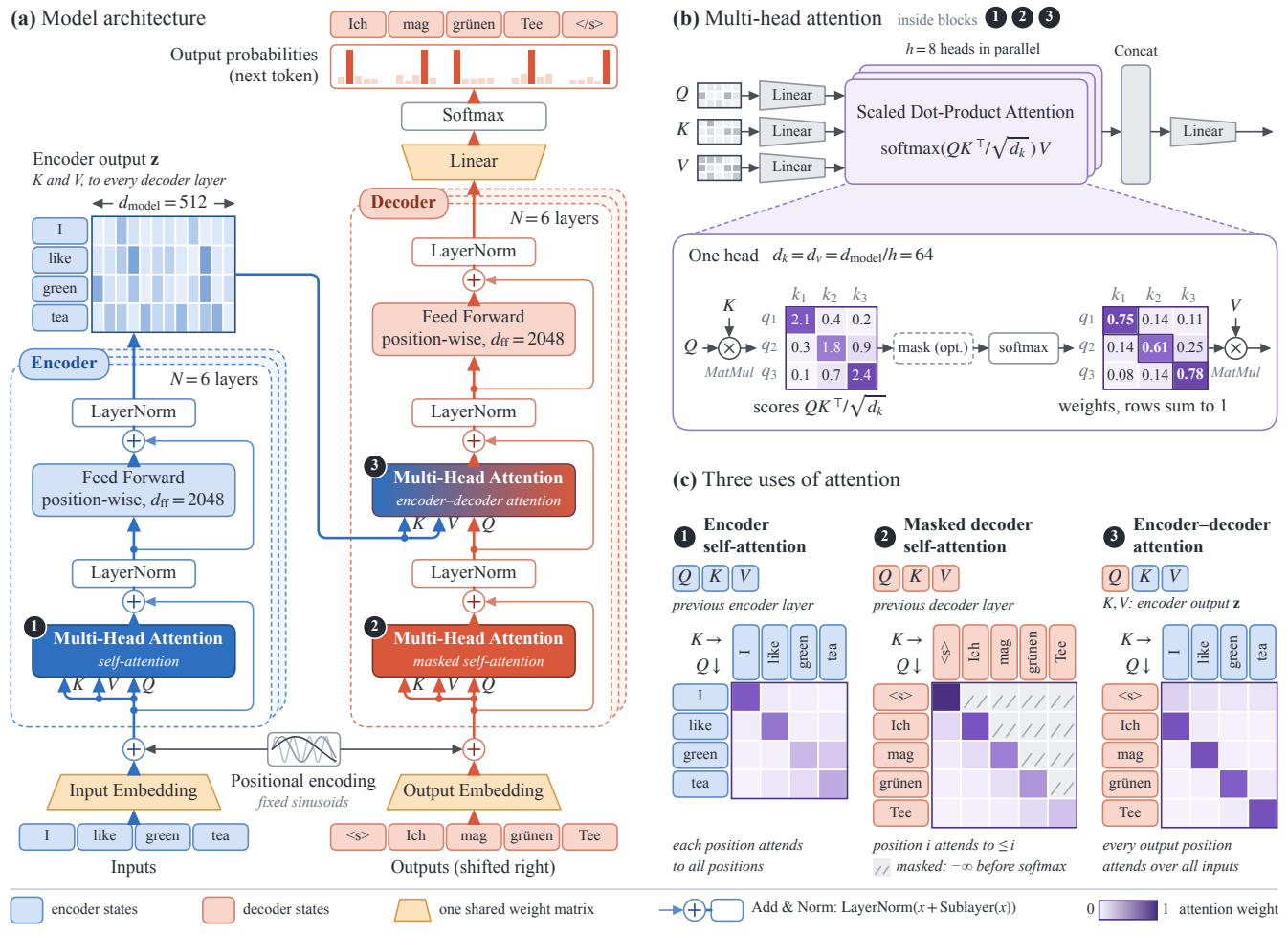

(a) Model architecture
Ich mag grünen Tee </s>
Output probabilities (next token)
Softmax
Linear
Decoder
N = 6 layers
LayerNorm
Feed Forward position-wise, d_ff = 2048
LayerNorm
3 Multi-Head Attention
encoder–decoder attention
K V Q
LayerNorm
2 Multi-Head Attention
masked self-attention
K V Q
LayerNorm
Output Embedding
<s> Ich mag grünen Tee
Outputs (shifted right)
Positional encoding
fixed sinusoids
Input Embedding
I like green tea
Inputs
Encoder output z
K and V, to every decoder layer
d_model = 512
I like green tea
Encoder
N = 6 layers
LayerNorm
Feed Forward position-wise, d_ff = 2048
LayerNorm
1 Multi-Head Attention
self-attention
K V Q
(b) Multi-head attention
inside blocks 1 2 3
h = 8 heads in parallel
Concat
Q Linear
K Linear
V Linear
Scaled Dot-Product Attention
softmax(QK^T/√d_k) V
Linear
One head d_k = d_v = d_model/h = 64
K
Q
MatMul
k1 k2 k3
q1 2.1 0.4 0.2
q2 0.3 1.8 0.9
q3 0.1 0.7 2.4
mask (opt.)
softmax
scores QK^T/√d_k
V
MatMul
k1 k2 k3
q1 0.75 0.14 0.11
q2 0.14 0.61 0.25
q3 0.08 0.14 0.78
weights, rows sum to 1
(c) Three uses of attention
1 Encoder self-attention
Q K V
previous encoder layer
2 Masked decoder self-attention
Q K V
previous decoder layer
3 Encoder–decoder attention
Q K V
K, V: encoder output z
K→ Q↓
I like green tea
each position attends to all positions
<s> Ich mag grünen Tee
position i attends to ≤ i
masked: −∞ before softmax
every output position attends over all inputs
encoder states
decoder states
one shared weight matrix
Add & Norm: LayerNorm(x + Sublayer(x))
0 1 attention weight